

CLEARWATER BEACH PIER 60, FL, USA

WMO: 997345

Lat: 27.9777N

Lon: 82.8321W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 42.1	(c) 45.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 32.2	(k) 56.6	(l) 29.2	(m) 55.6	(n) 12.3	(o) 60	(p) 0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 6.9	(c) 88.3	(d) N/A	(e) 87.2	(f) N/A	(g) 86.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.9	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 29.0	(b) 25.0	(c) 21.6	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 29.0	(b) 25.0	(c) 21.6	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.7	59.8	62.6	66.3	71.5	76.6	81.0	82.1	83.1	81.0	76.0	67.6	64.2
	DBStd	9.41	7.67	7.35	5.90	3.93	3.17	2.19	1.89	1.63	2.16	4.78	5.63	7.42
	HDD50	26	16	5	0	0	0	0	0	0	0	0	0	5
	HDD65	519	190	118	58	6	0	0	0	0	0	4	38	105
	CDD50	8313	321	359	506	644	824	929	995	1026	929	806	528	446
	CDD65	3331	29	52	99	201	359	479	530	561	479	344	116	82
	CDH74	29593	29	61	104	563	2469	5274	6225	6894	5014	2549	275	136
	CDH80	8324	0	4	3	26	262	1470	2125	2628	1408	383	13	2
(14) Wind	WSAvg	10.2	10.8	10.3	11.8	11.4	10.4	10.4	8.9	9.1	9.2	9.7	10.2	10.2
(15) Precipitation	PrecAvg	51.3	2.6	2.7	3.1	2.5	2.5	6.9	8.5	8.7	7.0	2.8	1.7	2.4
	PrecMax	71.3	5.4	11.2	8.8	8.6	8.3	13.9	15.0	13.8	13.7	6.9	5.5	15.4
	PrecMin	40.8	0.4	0.5	0.4	0.4	0.2	3.0	4.1	4.8	2.0	0.6	0.4	0.6
	PrecStd	7.8	1.5	2.2	2.4	1.7	2.0	3.0	2.5	2.7	3.1	1.4	1.2	3.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	79.3	79.7	82.9	86.4	89.1	89.3	90.1	90.1	87.0	82.2	79.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	74.4	75.6	76.8	80.3	84.1	86.9	87.2	88.0	87.8	84.8	79.6	77.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	72.1	73.8	75.0	78.5	82.5	86.1	86.4	87.2	86.2	83.6	77.6	75.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	70.1	71.8	73.6	77.1	81.4	85.2	85.9	86.6	85.2	82.2	75.5	73.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	9.5	9.4	8.4	7.2	6.5	6.7	6.9	8.2	8.9	10.0	10.4
		MCDBR	12.5	10.3	10.8	11.2	10.2	8.5	7.7	7.8	10.2	9.7	11.6	10.7
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.345	0.358	0.369	0.393	0.424	0.449	0.473	0.455	0.423	0.390	0.355	0.345
	taud		2.528	2.499	2.455	2.353	2.324	2.332	2.272	2.335	2.391	2.468	2.512	2.553
	Ebn at Noon		285	290	292	287	276	268	261	266	272	276	279	280
	Edh at Noon		29	32	35	40	41	41	43	40	37	32	29	27
(44) All-Sky Solar Radiation	RadAvg		1116	1347	1718	2066	2201	2000	1889	1810	1641	1514	1226	1030
	RadStd		82	95	152	106	117	159	112	112	118	95	71	86

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-7	-86	+301	+207

Nomenclature: See separate page